

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26183
- **Problem Statement Title** – *Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics*
- **Theme** – *Blockchain & Cybersecurity*
- **PS Category** – *Software*
- **Team ID** – BC-07
- **Team Name (Registered on portal)** – *FineX*



THE PROBLEM A victim reports one wallet. The money has already moved through several more, and the exchange that received it — the only party who can freeze it — is not named anywhere on the chain.

Manual tracing takes hours per case, and the freeze window is shorter than that.

OUR SOLUTION

FineX turns a victim-reported wallet — or the transaction hash their own exchange can produce — into the **customer deposit account inside the receiving exchange**, and states whether the money can still be reached.

- **It names the account, not just the exchange.** The deposit address an exchange can actually restrain — 241 of them derived from public data alone.
- **Triage, not just tracing.** Every complaint returns CRITICAL (funds still at rest), SUSPICIOUS (a freezable exit named) or CLOSED (mixer or sanctioned), so a cyber cell knows where the next hour goes.
- **Explainable by design.** Six behavioural rules, each stating in plain words why it fired. No model names an exchange or sets a disposition.
- **Complaints that share an account are one case.** Three of our ten recorded cases meet at a single MEXC deposit address.
- **Evidence, not a screenshot.** Each case produces a packet carrying the SHA-256 of every blockchain response, plus a drafted restraint request.

WHAT AN OFFICER PUTS IN

A wallet address, a transaction hash, or a pasted column of either — singly or in bulk.

WHAT COMES BACK

A named exit account · a disposition · ranked next wallets · an evidence packet · a freeze request.

CRITICAL

Funds still at rest — act now

SUSPICIOUS

Freezable exit named — request the restraint

CLOSED

Mixer or sanctioned — document and close

Working deployment, not a mock-up — every figure on these slides is counted from the committed repository.

TECHNOLOGIES USED

▪ **Next.js 16.3.4 · React 19.2.8 · TypeScript 5**

interface and API in one deployable

▪ **Tailwind CSS 4 · Node.js 22**

no component library, no build sprawl

▪ **TronGrid public API, native fetch**

TRON mainnet, USDT TRC-20

▪ **JSON datasets in the repository**

no database to provision or lose

▪ **Rule-based scoring**

no ML model, no language model, anywhere

▪ **SHA-256 chain-of-custody**

every chain response hashed into the packet

ONE ADDED DEPENDENCY

@xyflow/react for the flow canvas. Every other visual is hand-drawn SVG.

PIPELINE — AND THE LIMITS IT RUNS UNDER

1 Intake Wallet or 64-hex transaction hash · base58check validated before any chain read

2 Collection Paginated TRC-20 transfers · per-address cache · shared adaptive pacing (250 ms, 100 ms keyed)

3 Traversal BFS: depth ≤ 3 · top 5 outflows per wallet · drop $< 1\%$ of the amount · only what left after the money arrived

4 Taint Haircut (default) and FIFO, stated side by side · dwell measured from a real receipt

5 Attribution Deterministic lookup, priority sanctioned $\rightarrow$ hot wallet $\rightarrow$ deposit address · confidence + source tier on every label

6 Clustering ≥ 2 sweeps into one tagged wallet and $\geq 90\%$ of inflow forwarded $\Rightarrow$ customer deposit address (241 · 10 exchanges · 15 seeds)

7 Scoring SHORT_DWELL · HIGH_FANOUT · PEEL_CHAIN · ROUND_AMOUNTS · NEW_ADDRESS · SANCTIONED_CONTACT

8 Triage CRITICAL / SUSPICIOUS / CLOSED with a one-sentence reason, plus ranked leads and chokepoints

9 Output Evidence packet (hash per response) · freeze request · CSV worklist · watchlist

Progress is streamed per hop as the trace runs, so the officer watches real telemetry — not a fake progress bar.

FROM THE LIVE DEPLOYMENT — the named customer deposit address, its confidence and its evidence tier

CUSTOMER DEPOSIT ADDRESS

TX1so33jdGd8JkYD7JVB6q1i4QUDhPB2MN  Copy

Likely MEXC deposit cluster · 95% confidence **HEURISTIC**

If confirmed by MEXC, this is the account that can be frozen. Name it in the request — not just the exchange.

EVERY RISK, PAIRED WITH WHAT WE ACTUALLY DO ABOUT IT

RISK Public API rate limits

MITIGATION One shared adaptive pacer for the whole server — 250 ms between calls, 100 ms with a key, doubling on every refusal. A wallet we could not read is reported as unreadable, never as empty.

RISK Mixers and bridges end the trail

MITIGATION We record the entry and stop. The case closes as CLOSED. We never guess past a bridge.

RISK Attribution is a heuristic, not proof

MITIGATION Every label carries a confidence and an evidence tier. Re-tested against the chain: 40 sampled, 31 readable, all 31 still meet the rule. Only the exchange can confirm the accountholder — the packet says so.

RISK Some behavioural rules are weak

MITIGATION Measured, not assumed: on a sample of 17 unreported wallets, peel-chain fired on 16 and fan-out on 15, so both are known to be weak on their own.

RISK No Indian VASP is publicly tagged

MITIGATION 2,500 tagged accounts scanned and none found. We do not invent one; adding a verified Indian exchange is one row of data and one re-run of the clustering.

RISK Cost, deployment and data custody

MITIGATION Public endpoints, no licence, no database. Self-hosted by the agency, so case data never leaves it. ₹0 in data licences as deployed.

MEASURED, FROM THE COMMITTED REPOSITORY

241

customer deposit
addresses derived

10

exchanges covered,
from 15 public seeds

202

OFAC-sanctioned
TRON addresses

6 / 6

behavioural rules fire
on real recorded cases

33 / 0

transactions confirmed
on re-read / mismatched

₹0

in commercial
data licences

WHAT IT CHANGES

- **For the investigator.** A named account to act on and a disposition that says which case is worth the next hour.
- **For the exchange.** A restraint request that names the deposit address, not just the brand — the account they can actually locate.
- **For the case file.** Findings re-verifiable from the hash of every chain response, months later.
- **For the state.** Sovereign and self-hosted: public data only, no foreign vendor, no per-seat licence.

Mixers: a path touching a sanctioned service closes the case. A dedicated mixer list is deliberately empty — no citable public source, and a wrong entry would close a case that should stay open.

AGAINST THE PROBLEM STATEMENT

BUILT NOW

VASP identification · wallet tracing · real-time analysis · fund-flow visualisation · exchange clustering · pattern detection · standard reports · investigative recommendations · dashboards · complaint intake · API integrations · wallet risk categorisation · automated alerts

AT THE FINALE

Cross-chain adapter with USDT on Ethereum end to end · NCRP intake API returning a restraint request against an acknowledgement number · server-side continuous monitoring

AFTER SIH

Live NCRP / SAHYOG connection once I4C grants access · self-hosted TRON node for indexing at scale · ML that ranks the queue only — never names an exchange or sets a disposition

1. Smart India Hackathon 2026 — Problem Statement SIH26183, Ministry of Home Affairs / I4C
<https://www.sih.gov.in/sih2026PS>
2. UNODC (2024) — Casinos, Money Laundering, Underground Banking and Transnational Organized Crime in East and South-East Asia, p. 20: USDT on TRON “has become a preferred choice”. This is why the tool is TRON-first.
https://www.unodc.org/roseap/uploads/documents/Publications/2024/Casino_Underground_Banking_Report_2024.pdf
3. OFAC Specially Designated Nationals list, US Department of the Treasury — source of the 202 sanctioned TRON addresses carried in the tool
<https://sanctionslist.ofac.treas.gov/Home/SdnList>
4. TronGrid API documentation, TRON — TRC-20 transfer data used for every trace
<https://developers.tron.network/reference/trongrid-v1-api-overview>
5. NCRP and SAHYOG — the platforms the problem statement asks this system to integrate with
<https://cybercrime.gov.in> · <https://sahyog.mha.gov.in>



Source code and documentation github.com/reemrasheed2007/crypto-fraud-tracer

Live deployment crypto-fraud-tracer.onrender.com



crypto-fraud-tracer.onrender.com

Three recorded cases open straight from the front page.

DEMO VIDEO — 58 SECONDS

A full case, start to finish, with narration.

paste the video link here

KEY RESEARCH INSIGHT

Public blockchain data → traceable fund movement → a named account an exchange can freeze.